

PDE - Problem Sheet

Course Code: MSMAT03DSC12

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1 Method of Characteristics

- Find and sketch some sample characteristic curves of the PDE:

$$(x+2)u_x + 2yu_y = 2u$$

in the $x-y$ plane. Write the ODE for u along a characteristic curve with x as the parameter and then solve the PDE with the initial condition $u(-1, y) = \sqrt{|y|}$, $y > 0$.

- Consider the PDE $xu_x + yu_y = 2u$, $x > 0, y > 0$. Plot the characteristic curves and solve the equation with the following initial conditions in the domain given above:

- $u = 1$ on the hyperbola $xy = 1$.
- $u = 1$ on the circle $x^2 + y^2 = 1$.
- Can you solve the equation, in general, if certain initial data is prescribed on the initial curve $y = e^x$? Justify with reasons.

- Sketch the characteristic curve, the initial curve, and solve the following problems:

- $xu_x + yu_y = ku$, $x \in \mathbb{R}$, $y \geq \alpha > 0$; $u(x, \alpha) = F(x)$, where k, α are fixed and F is a given smooth function.
- $yu_x - xu_y = 0$; $u(x, 0) = x^2$.
- $x^2u_x - y^2u_y = 0$; $u(1, y) = F(y)$.
- $yu_x + xu_y = 0$; $u(0, y) = e^{-y^2}$.

- Solve the quasi-linear problem and verify transversality conditions:

- $uu_x + u_y = 0$, $u(x, 0) = x$.
- $uu_x + u_y = 1$, $u(x, x) = x/2$, $x \in (0, 1]$.

- Consider the PDE $u_x + u_y = 0$ on $\mathbb{R}^2$. Discuss the existence and uniqueness of the solution under the following initial conditions:

- $u(x, -x) = x$
- $u(x, x) = x$
- $u(x, x) = 1$

- Solve the PDE: $uu_x + u_y = 1$ with initial conditions $x = s$, $y = s$ and $u = s/2$ with $0 \leq s \leq 1$.

- Consider the equation $p^2 + q^2 = 1$ with initial condition $u(x, y) = 0$ on the line $x + y = 1$. Find the two solutions using the method of characteristics.

- Find a smooth function $a : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that, for the equation of the form

$$a(x, y)u_x + u_y = 0,$$

there does not exist any solution in the entire $\mathbb{R}^2$ for any nonconstant initial value prescribed on $\{y = 0\}$.

9. Consider the PDE $xu_x + yu_y + zu_z = 3u$ in $\mathbb{R}^3$.
- (a) Solve the PDE with initial condition $u(x, y, 1) = x^2 + y^2$.
 - (b) Is it possible to find unique solution if the initial condition is prescribed on the surface $z = 1 + x^2 + y^2$?
10. Consider the following IVPs:
- A. $u = u_x^2 - 3u_y^2$, $u(x, 0) = x^2$, $x > 0$.
 - B. $u = u_x u_y$, $u(x, 0) = x^2$, $x > 0$.
- (a) Discuss the existence and uniqueness of both IVPs.
 - (b) Solve any one the above.
11. Consider the PDE $xu_x + yu_y = 2u$ on $\mathbb{R}^2$. Discuss the existence and uniqueness of the solution (both global and local) to the following initial conditions. If unique solution does not exist, find an alternative solution on $\mathbb{R}^2$.
- (a) $u = 1$ on the hyperbola $xy = 1, x > 0$.
 - (b) $u = 1$ on the line $y = 1$.
 - (c) $u = 1$ on the circle $x^2 + y^2 = 1$.
 - (d) $u(x, e^x) = xe^x$, for all $x \in \mathbb{R}$.